

Emit compile-check triage — full-corpus independent-witness sweep (2026-07-16)

What this is. A one-time broad sweep compiled EVERY `test/*.zbr + examples/*.zbr` (399 files) with the selfhost (`zebra.exe --emit-zig → zig build-exe -fno-emit-bin`), not just the ~201 smoke-registered files that `tools/compile_check.sh` normally covers. Goal: surface latent emit miscompiles (the BUG-182 class) hiding in un-gated files.

Headline: 293 PASS / 52 FAIL / 54 SKIP. The 52 FAILs triage below. This was found while hunting the generalized BUG-182 (dedicated-`Type_-variant`) class; BUG-182 was the visible tip of a much larger un-gated emit surface.

Honest caveats (do not over-read the "52"):

- It is NOT "52 bugs." ~11 are expected-fail (negative tests) or harness limits.
- Of the ~41 remaining, buckets are error SIGNATURES, spot-checked as real on current APIs (`Http`, `CsvWriter`, `Math` all verified current), but NOT every file was individually confirmed real-vs-stale. Some may be stale tests using evolved APIs.
- Many files share a ROOT CAUSE (e.g. 5 GUI files, one `param shadows 'init'` bug), so the real work is ~15–20 root causes, not 40 fixes.

Status legend: `NEG` expected-fail test · `HARNESS` external dep/platform · `REAL?` real-looking, unverified · `REAL✓` confirmed · `FIXED` ·

A. Negative / error-detection tests (expected to fail) — ~8, not bugs

(Some emit broken Zig instead of a clean TC error — a minor quality gap, not a miscompile.)

File	Error	Status
field_not_found_test	no field 'y' in P	NEG (registered smoke-negative)
method_not_found_test	no method 'shout' in []const u8	NEG (registered)
forgot_parens_test	value of type 'fn() void' ignored	NEG (compile_check SKIP)
bug108_this_outside_class_test	undeclared 'self'	NEG (<code>this</code> outside class)
bug106_heterogeneous_list_test	expected i64, found *const [3:0]u8	NEG? (heterogeneous list)
bug105_enum_member_test	expected i64, found Color	NEG? (verify)
bug105_union_variant_test	expected i64, found union tag	NEG? (verify)

File	Error	Status
bug099_unresolved_test	undeclared 'Math'	NEG? (unresolved-detection; Math IS current)

B. Harness limits (external deps / platform) — 3, not bugs

File	Error	Status
c_interop_test	unable to load 'CUtils.zig' (FileNotFound)	HARNESS (external C)
zig_interop_test	unable to load 'ZigMath.zig' (FileNotFound)	HARNESS (external Zig)
plugin_host	unsupported platform (dynamic_library)	HARNESS (platform)

C. Preamble `reduce` param shadows top-level `init` — **FIXED (BUG-184)** — cleared 3 files

The preamble helper `_zebra_list_reduce` had a parameter `init` that shadowed a user top-level `pub fn init` (MVU model ctor). Renamed → `init_val`. **Cleared:** `counter` · `hbox_smoke` · `file_dialog_smoke`. The other 3 were failing on `init` FIRST; now reclassified to their next bug:

- `panel_smoke` → **new:** function parameter 'g' shadows function parameter from outer scope (nested closure param `g` shadowing the outer `g` — a distinct shadowing bug).
- `widget_smoke` → **D4** (`^T-box *T` vs `*const T`).
- `test_gui_simple` → local constant shadows declaration of 'frame' (const-shadow, akin to C but a local `frame` — likely a `with/capture-block` re-binding).

D. Genuine codegen / type miscompiles on CURRENT constructs — ~30, clustered

D1 — undeclared type not emitted (stdlib type used but not materialized) — REAL?

File	Undeclared	Note
csv_test	<code>CsvWriter</code>	<code>CsvWriter</code> is current (QUICKSTART, Resolver) → real emit gap
http_test	<code>Http</code>	<code>Http</code> is current (CodeGen 11597); client <code>.get</code> path?
https_test	<code>Http</code>	same
zebra_ide	<code>List</code>	<code>List</code> undeclared — investigate

D2 — SYNTAX errors in emitted Zig (malformed emit) — REAL✓ (Zig can't even parse it)

File	Error	Status
json_test	expected ';' after statement	root cause FIXED (BUG-183) — single-quoted string with embedded " emitted unescaped. <code>json_test</code> still fails on a <i>separate</i> D7 bug (never-mutated).

File	Error	Status
expose_dotted_test	expected ';' after declaration	OPEN — likely a <i>different</i> malformed emit (verify; not the string-escape cause)
selfhost_probe6	expected ';' after initializer	OPEN — likely a different malformed emit (verify)

BUG-183 (FIXED 2026-07-16): single-quoted string literals with an embedded literal `"` emitted unescaped Zig (`'{"a":1}'` → `"{"a":1}"` → syntax error). Selfhost-only (bootstrap was correct). Fixed via idempotent escaping in `escapePlainStr` ([selfhost/CodeGen.zbr](#)). This is a *class* fix — any single-quoted string with embedded quotes (JSON/HTML/etc.) across the corpus.

D3 — `.len/.member` on `ArrayList` / str-vs-list confusion (`StrSet`/BUG-182 family) — REAL?

File	Error
dns_test	no field 'len' in <code>ArrayList([]const u8)</code>
tc_types_test	struct 'TcTypes' has no member 'len'
string_methods_test	expected <code>[]const u8</code> , found <code>ArrayList([]const u8)</code>
crossmod_expose_test	<code>ArrayList(Tag)</code> is not indexable

D4 — `^T`-box pointer mismatch (`*T` vs `T` / `*const T`) — REAL?

File	Error
generic_pair_test	expected 'T', found '*T'
method_chain_throws_test	expected '*T', found '*const T'
selfhost_probe5	expected '*Expr', found 'Expr'
tc_check_test	expected '*TcExpr', found 'TcExpr'
tc_infer_test	expected '*TcExpr', found 'TcExpr' (same as tc_check)

D5 — method on a mis-typed receiver (BUG-182 class) — REAL?

File	Error
datetime_test	no method 'toEpoch' in <code>_DateTime</code>
tcp_advanced_test	no method 'write' in <code>***TcpConn</code> (optional not unwrapped)
extend_test	no method 'shout' in <code>[]const u8</code> (extension method)
json_parse_typed_test	no method 'getString' in <code>json.dynamic.Value</code>
fuzzy_match	no method 'concat' in <code>[]u8</code>
fuzzy_selfhost	no method 'concat' in <code>[]u8</code> (same as fuzzy_match)

D6 — argument-count mismatch — REAL?

File	Error
expressiveness_test	member function expected 2 arg(s), found 1
progress_test	expected 2 argument(s), found 1

D7 — emitted Zig trips Zig strictness (const/var, unused, unreachable) — REAL? (lower severity)

File	Error
gui_test	local variable is never mutated (var should be const)
unicode_test	local variable is never mutated
typechecker_test	unused function parameter
file_io_test	unreachable code

D8 — misc singletons — REAL?

File	Error
derive_test	operator == not allowed for type 'Point' (@derive == gap?)
reflect_test	missing struct field: name
escape_field_test	missing struct field: items
log_test	expected 'u8', found '*const [5:0]u8' (char vs string)
math_test	atan2 not implemented for comptime_float (emit should coerce to runtime)
raise_details_test	error union is ignored
throws_autoprop_test	error union is ignored (same signature)

Recommended campaign (for 1.0)

1. **Verify each D-cluster real-vs-stale** (spot-checks say real; confirm the rest). Sean's context on which tests are live vs abandoned will sharpen this fast.
2. **Fix by root cause, not by file** — C is 1 fix for ~6 files; D4/D5 share mechanisms (^{^T} boxing, mis-typed-receiver dispatch — the BUG-182 family). Est. ~15–20 root causes.
3. **Gate compile-check over the triaged-clean corpus** (see CLAUDE.md "Verification gates"; currently manual) so this surface cannot silently regrow.
4. Prune genuinely-stale tests (quarantine, don't delete — note why).

Raw per-file error capture: [scratchpad/errors.txt](#) (session artifact) — regenerate via the broad sweep in [tools/compile_check.sh](#) extended to the full corpus.